

Name: → Anirui Podder

Date: → 22.1.22

B.Tech CSE(AIMI)-C Student Codes - BWU/BTA/21/176

EXPERIMENT No.02

Title:-

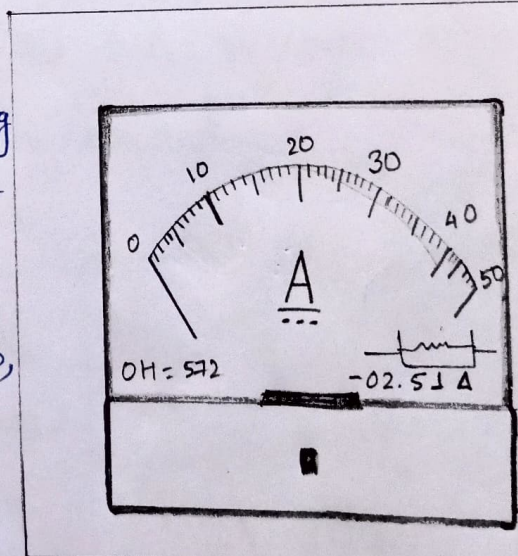
Introduction and uses of following instruments (a) Voltmeter
(b) Ammeter (c) Multimeter (d) Oscilloscope.

Demonstration of physical resistors, capacitors with Color Code, inductors and auto-transformers.

Objective:- To be familiarized with electrical measuring instrument and other components.

Ammeter:-

- Ammeter is employed for measuring of current in a circuit and connected in series in the circuit.
- As ammeter is connected in series, the voltage drop across ammeter terminals is very low.
- This requires that the resistance of the ammeter should be as low as possible.
- The current coil of ammeter has low current carrying capacity whereas the current to be measured may be quite high.



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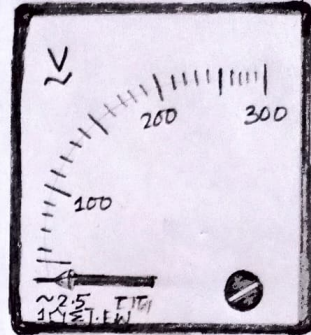
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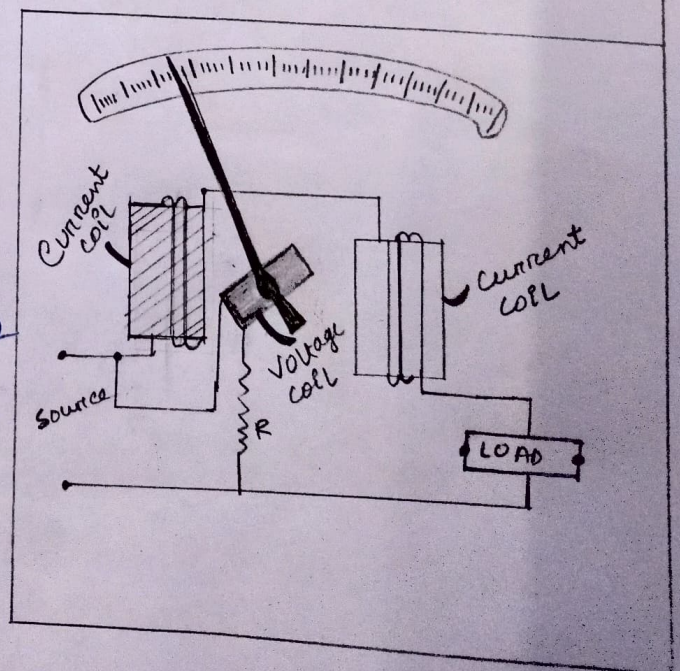
VOLTMETER:-

- Voltmeter is employed to measure the potential difference across any two points of a circuit.
- It is connected in the parallel across any element in the circuit.
- The resistance of Voltmeter is kept very high by connecting a high resistance in series of the Voltmeter with the current coil of the instrument.
- The actual Voltage drop across the current coil of the Voltmeter is only a fraction of the total Voltage applied across the Voltmeter which is to be measured.



WATTMETER:-

- The measurement of real power in AC Circuits is done by using an instrument using Wattmeter. The real power in AC Circuits is given by expression $VI \cos \phi$, where $\cos \phi$ is power factor. A Wattmeter has two coils, namely Current coil and pressure coil.



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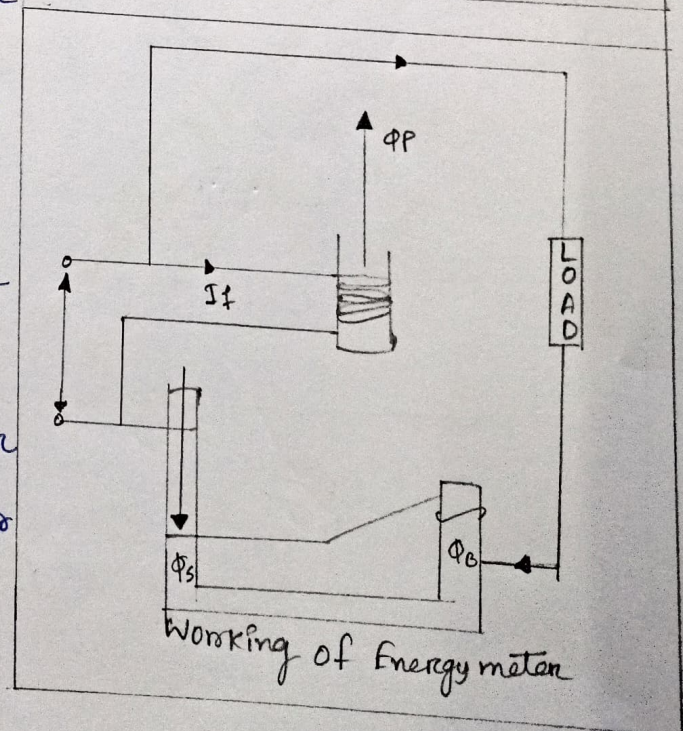
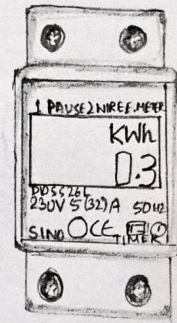
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- The Current Coil (CC) is connected in series with the load and the pressure coil (PC) is connected across the load.
- Wattmeters are available in dual range for voltages as well as for current.

ENERGY METER:-

- Energy meter is an instrument which is used to measure the consumption of electric energy in a circuit (DC or AC).
- It measures energy in kWh.
- The essential difference between a energy meter and a wattmeter is that the former is fitted with some type of registration, mechanism where by all the instantaneous readings of power are summed over a definite period of time whereas the latter indicates the value at particular instant when it is read.



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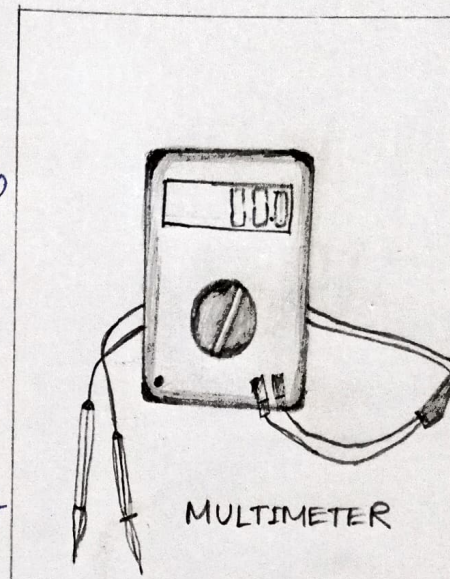
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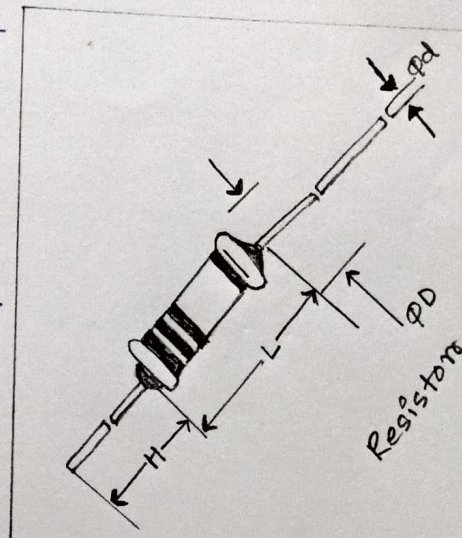
Multimeters:-

- Multimeter is a measuring instrument used to measure the current, voltage and resistance. These can be used to troubleshoot many electrical equipment such as domestic appliances, power supplies etc.



Resistor:-

- A resistor is a passive electronic component and is used to create resistance in the flow of electric current. In simple words, it limits the flow of electrons through the circuit.



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RESISTOR COLOR TABLE



	1st Digit	2nd Digit	Multiplication	Tolerance
BLACK	0	0	$\times 1\ \Omega$	$\pm 1\%$ $\pm 2\%$
BROWN	1	1	$\times 10\ \Omega$	
RED	2	2	$\times 100\ \Omega$	
ORANGE	3	3	$\times 1\text{K}\ \Omega$	$\pm 5\%$ $\pm 10\%$
YELLOW	4	4	$\times 10\text{K}\ \Omega$	
GREEN	5	5	$\times 100\text{K}\ \Omega$	
BLUE	6	6	$\times 1\text{M}\ \Omega$	$\pm 5\%$ $\pm 10\%$
VIOLET	7	7	$\times 0.1\ \Omega$	
GREY	8	8	$\times 0.01\ \Omega$	
WHITE	9	9		

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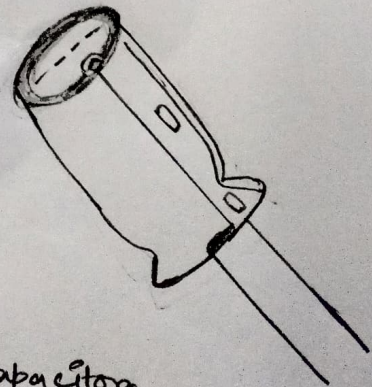
COLOUR CODING :-

COLOR	1 st BAND	
BLACK	0	→ B
BROWN	1	→ B
RED	2	→ R
ORANGE	3	→ O
YELLOW	4	→ Y
GREEN	5	→ GREAT
BLUE	6	→ BRITAIN
VIOLET	7	→ VER
GREY	8	→ GR
WHITE	9	→ W
GOLD	1	] → Tolerance Bands
SILVER		

Capacitors :-

- The capacitor is a passive component and it stores the electrical energy into an electrical field.

The effect of the capacitor is known as a capacitance. It is made up of two close conductors and separated by the dielectric material.



Capacitors

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If the plates are connected to the power then the plates accumulate the electric charge. one plate accumulates the positive charge and another plate accumulates the negative charge.

Different Types Of Capacitors:-

- Electrolytic Capacitors.
- Mica Capacitors
- Paper capacitors
- film capacitors
- Non-polarized Capacitors
- Ceramic capacitors

Types Of Inductors:-

- Air-Core Inductors
- Iron-Core Inductors
- Toroidal Inductors
- Laminated Core Inductors.
- Powdered Iron-Core Inductors.

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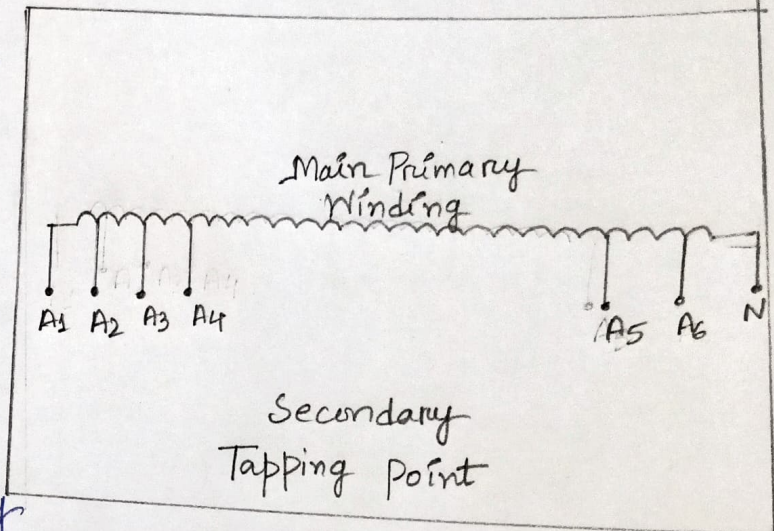
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Autotransformer:-

- It has only one single-voltage winding which is common to both sides.

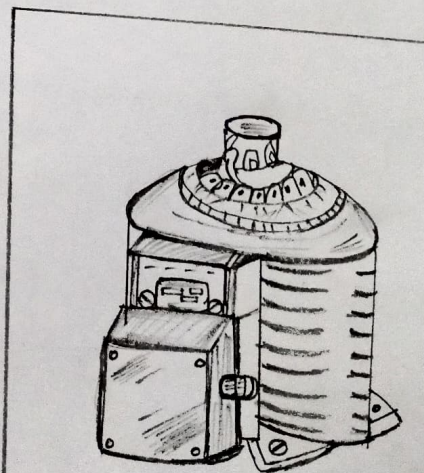
- The single winding is "tapped" at various points along its length to provide a percentage of the primary voltage supply across the secondary load.



- Then the autotransformer has the usual magnetic core but only has one winding, which is common to both the primary and secondary circuits.

VariaC:-

- The construction of a variable autotransformer, or variaC, is the same as for the fixed type. A single primary winding wrapped around a laminated magnetic core is used as in the autotransformer but instead of being fixed at some predetermined



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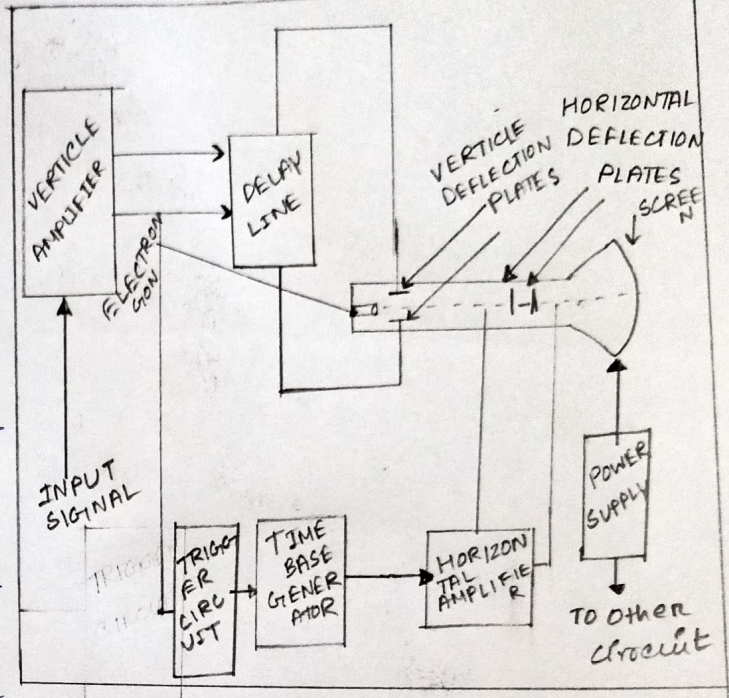
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OSCILLOSCOPE:-

A Cathode Ray Oscilloscope (CRO) is an instrument generally used in a laboratory to display, measure and analyze various waveforms of electrical circuits. A Cathode ray Oscilloscope plotter that can display an input signal versus time or another signal.



Conclusion:-

In this experiment, we can see many types of electrical instrument like ammeter, voltmeter, wattmeter, ~~mult~~ multimeter and so on. Ammeter measures the current, voltmeter measures voltage and wattmeter are used for measuring the power. By this we come to know about all types of electrical instrument and their uses. We ~~cap~~ are also familiarized with the capacitors, color coding of resistors, inductors, variac and also oscilloscope.

Exp NO. 02

Title:-

Introduction and uses of following instruments:

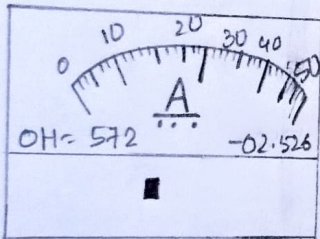
(a) Voltmeter (b) ammeter (c) Multimeter (d) Oscilloscope.

Demonstration of physical resistors, capacitors with color code, Inductors, and ~~auto~~ auto transformer.

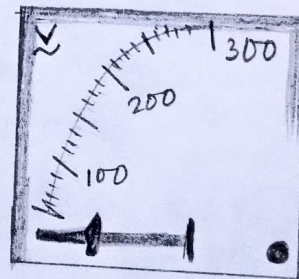
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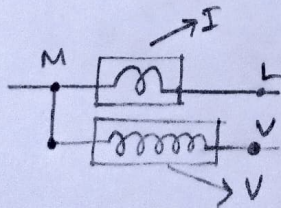
Ammeter



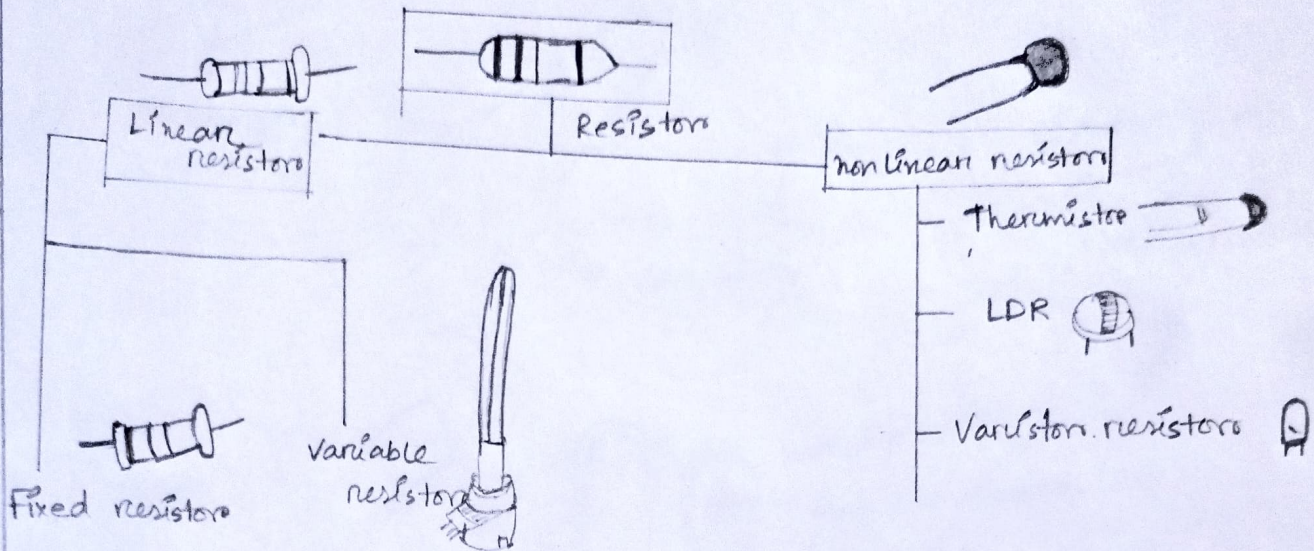
Volmeter



Wattmeter



Type of Resistors:-

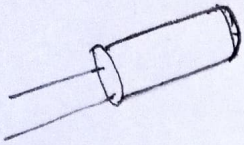


Resistors color table

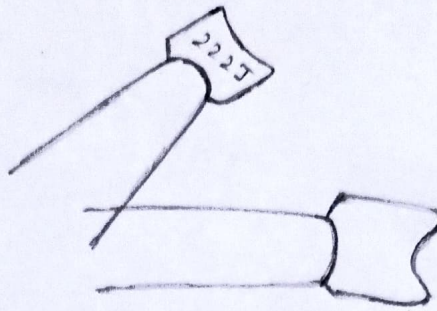


	1st digit ↓	2nd Digit ↓	Multiplicier ↓	Tolerance ↓
Black	0	0	X 1 Ω	+ 1%
Brown	1	1	X 10 Ω	± 1%
Red	2	2	X 100 Ω	± 2%
Orange	3	3	X 1 K Ω	
Yellow	4	4	X 10 K Ω	
Green	5	5	X 100 K Ω	
Blue	6	6	X 1 M Ω	
Violet	7	7		
Grey	8	8	X 0.1 Ω	± 5%
White	9	9	X 0.01 Ω	± 10%

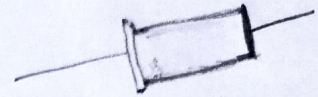
Capacitors



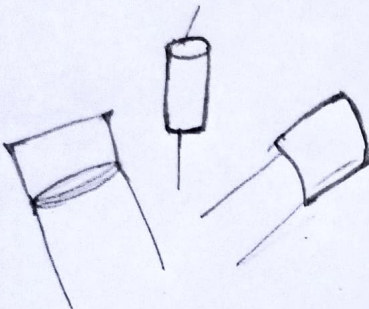
Electrolytic capacitor



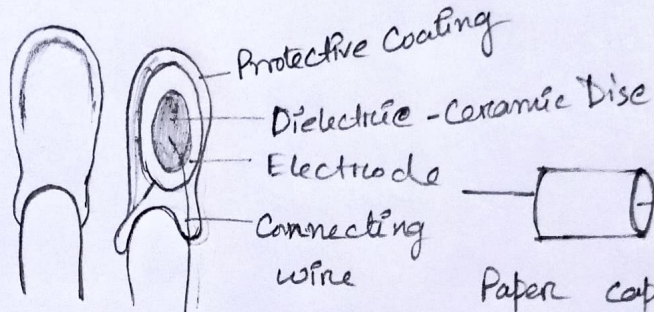
Mica capacitor



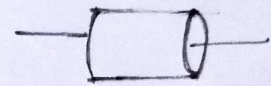
Non polarised



Film capacitor



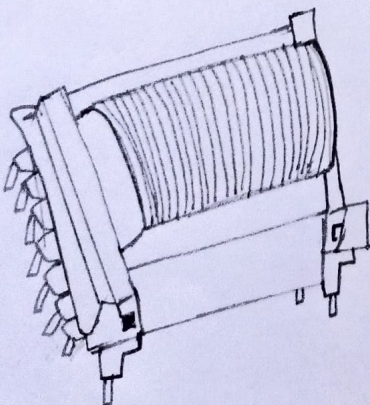
Ceramic capacitor



Paper capacitor

Inductors

Iron Core Inductors



Toroidal Inductors

